Open-science conventions as agent infrastructure

A stack-aware workflow experiment · *Proposed contribution to the LSM Future Skills 2026 breakout* — AI in Daily Research: Sharing What Works (and What Doesn't) · *22 September 2026*

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Most AI-in-research demos I come across focus on a single task — cleaning code, summarising a paper, drafting an email. I'd like to share something I've found surprisingly important: how *useful* an AI agent is on a research project depends less on the agent than on how the project itself is organised. When the project follows familiar open-science conventions, an agent can navigate it with very little hand-holding — read the right file, run the right pipeline, write its output back in the same structured form. The conventions do most of the work; the agent is a convenient interface to them.

The stack I rely on

Layer	Tool	Role
Data layout	**BIDS**	the directory *is* the API
Versioning	**DataLad** + git	data and code, with receipts
Pipelines	**Snakemake**	declarative, inspectable DAGs
Notebooks	**Marimo**	reactive cells; no hidden state; great for exploration
Publication	**Quarto / MkDocs**	one source → site, slides, book, reports
Agent permissions	**Model Context Protocol (MCP)**	scoped, auditable agent–tool surface

None of these is new — which is part of the point. What I find interesting is how they compose: treated as **one surface**, they turn into something an agent can navigate end-to-end.

A quick demo (≈ 4 min)

- One prompt → three linked documents (a short literature survey, a design spec, a roadmap) written by three agents in under 20 minutes.
- Each agent reads what the previous one wrote — **file-mediated handoff**, not ephemeral state. The audit trail back to source stays visible throughout.

What I haven't solved yet

Agents still write confident result-claims that aren't tied back to the executed notebook cell that produced them, so I end up re-running analyses by hand to check provenance. I'd be curious to hear how others in the room handle this.

Format

≈ **10 min talk + discussion** (very flexible to whatever fits the session best). I'll demo one open-source implementation I've been working on — projio — but the takeaway I'd most like to land is the *convention work itself*, which is portable to whatever stack you already use. Attendees would leave with a small starter repository and one MCP query they can try against their own project.

This also doubles as a soft preview for a 4-day follow-up workshop on the same stack I'm planning at LMU in late September 2026 (PhDs and postdocs welcome). Happy to share more if it's useful.